

Module 8: Introduction to Probability

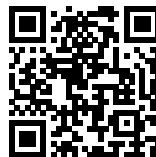
Instructional Hours: 4

Hello, you are now at the first part of this module. You can begin by answering the following questions to measure your prior knowledge on Probability.

Probability Terms and Axioms

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/4ewDPUPApca>



Most often you rely on probability to make decisions in your daily life, for instance, by using your own estimates to predict what could happen. The word 'probability' originates from the word 'probable', which means possibility. Also, possibility depends on the chance factor. Probability theory is the mathematical modeling of the phenomenon of chance or randomness. It is also known as that branch of mathematics that constructs, analyzes and compares probability models of random phenomena. In simple terms, probability can be defined as the quantification of uncertainty or chance. In order to understand probability, you need to remember that when an experiment takes place it can result in a variety of outcomes. These outcomes may be referred to individually or in a combination as an event.

Definitions of Terms

- **Probability** - This is the measure of the likelihood that an event will happen and it is given on a numerical scale from 0 to 1 . The numbers representing probability can be written as percentages, fractions or decimals. Whenever we are given a probability of 0 , this indicates that that event is impossible. In addition, when the probability is 1, this indicates that the event is certain to happen. All other events have a probability between 0 and 1.
- **Random Experiment:** - This is an experiment that, at least theoretically, may be repeated as often as we want and whose outcome cannot be predicted. Note that the outcomes are different even though the experiment is repeated in the same manner every time. The typical examples of random experiments include but are not limited to roll of a die, the toss of a coin, drawing a card from a deck and many more. To model and analyze an experiment, we must understand the set of possible outcomes from the experiment.

- **Sample Space:** - A Sample Space, S , is the set of all logically possible outcomes resulting from some experiment. An element of a sample space is called a sample point or more simply an outcome, and is denoted by s .

Example of sample spaces are

1. If the outcome of an experiment consists in the determination of the sex of a new born child, then

$$S = \{g, b\}$$

2. If the experiment consists of running a race among the seven horses having post positions 1, 2, 3, 4, 5, 6, 7, then

$$S = \{ \text{all ordering of } (1, 2, 3, 4, 5, 6, 7) \}$$

3. Suppose we are interested in determining the amount of dosage that must be given to a patient until that patient reacts positively. One possible sample space for this experiment is to let S consist of all positive numbers. That is, let

$$S = (0, \infty)$$

- **Event:** - An event A is a subset of the sample space of a random experiment. That is, $A \in S$. The event that contains no sample points at all is called an impossible event and is denoted by $\emptyset$. In modelling an experiment, we can also be interested in describing or formulating new events by having a combination of different events. In this case we use the concepts of sets and operations on sets.
- The union of two events is the event that consists of all outcomes that are contained in either of the two events. In symbols, given two events A and B , then their union is $A \cup B$.
- The intersection of two events is the event that consists of all outcomes that are contained in both of the two events. In symbols, given two events A and B , then their intersection is $A \cap B$.
- The complement of an event in a sample space is the set of outcomes in the sample space that are not in the event. In symbols, the complement of an event A is A' .

Example 1. *Samples of polycarbonate plastic are analyzed for scratch and shock resistance. The results from 50 samples are summarized below*

	<i>Shock Resistance</i>	<i>high</i>	<i>low</i>
<i>Scratch Resistance</i>	<i>high</i>	40	4
	<i>low</i>	1	5

Let A denote the event that a sample has high shock resistance, and let B denote the event that a sample has high scratch resistance. Determine the number of samples in $A \cap B$, A' and $A \cup B$.

Solution: The event $A \cap B$ consists of the 40 samples for which scratch and shock resistances are high. The event A' consists of 9 samples in which the shock resistance is low. The event $A \cup B$ consists of the 45 samples in which the shock resistance, scratch resistance, or both are high.

Empirical Probability

This interpretation of probability relies on actual experience to determine the likelihood of outcomes. The formula is as follows,

Given a frequency distribution, the probability of an event A being in a given class is

$$P(A) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}}$$

Example 2. In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood and 2 had type AB blood. Set up a frequency distribution and find the following probabilities

- (i) A person has type O blood
- (ii) A person has type A or type B blood
- (iii) A person has neither type A nor type O blood
- (iv) A person does not have type AB blood

Solution: The frequency distribution is given as

Type	Frequency
A	22
B	5
AB	2
O	21

- (i) $P(O) = \frac{21}{50}$
- (ii) $P(A \text{ or } B) = \frac{22}{50} + \frac{5}{50} = \frac{27}{50}$
- (iii) $P(\text{Neither } A \text{ nor } O) = \frac{2}{50} + \frac{2}{50} = \frac{4}{50} = \frac{2}{25}$
- (iv) $P(\text{not } AB) = 1 - P(AB) = 1 - \frac{2}{50} = \frac{48}{50} = \frac{24}{25}$

Subjective Probability

Subjective probability uses a probability value based on an educated guess or estimate, employing opinions and inexact information. This guess is often based on the person's experience and evaluation of a situation. For example, a physician may say that on the basis of her diagnosis, there is a 30% chance the patient will need an operation.

You are making right steps, you are now going to look at axioms of probability.

Probability Axioms

To discuss the axioms of probability, we denote events by capital letters and write the probability of an event A occurring as $P(A)$. The following axioms only apply to discrete sample spaces - those with only finite (or countably finite) sets of outcomes.

1. $P(A) \geq 0$; The probability of an event is a non-negative real number.
2. $P(S) = 1$
3. For any number of mutually exclusive events, $A_1, A_2, A_3, \dots$ we have

$$P(A_1 \cup A_2 \cup A_3 \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

Remark. *These three axioms do not tell us how to assign probabilities to events; they merely restrict the ways in which it can be done.*

Example 3. *An experiment has four possible outcomes, A , B , C , and D , that are mutually exclusive. Explain why the following assignments of the probabilities are not permissible:*

(i) $P(A) = 0.12, P(B) = 0.63, P(C) = 0.45, P(D) = -0.20$

(ii) $P(A) = \frac{9}{120}, P(B) = \frac{45}{120}, P(C) = \frac{27}{120}, P(D) = \frac{46}{120}$

Solution: (i) $P(D) = -0.20$ violates the first axiom

(ii) $P(S) = P(A \cup B \cup C \cup D) = \frac{9}{120} + \frac{45}{120} + \frac{27}{120} + \frac{46}{120} = \frac{127}{120} \neq 1$, and this violates the second axiom.

Probability Rules

Addition Rules

Based on the three axioms of probability, we can derive many other rules that have important applications. For example, the probability of a joint event can often be determined from the probabilities of the individual events that comprise it. Basic set operations are also sometimes helpful in determining the probability of a joint event.

1. If A and A' are complementary events in a sample space S , then

$$P(A') = 1 - P(A)$$

We know that if two events are mutually exclusive, then $A \cup A' = S$, therefore

$$\begin{aligned} 1 &= P(S) \quad \text{by axiom 2} \\ &= P(A \cup A') \\ &= P(A) + P(A') \end{aligned}$$

and it follows that $P(A') = 1 - P(A)$

2. $P(\emptyset) = 0$ for any sample space S .

Since S and $\emptyset$ are mutually exclusive and $S \cup \emptyset = S$ in accordance with the definition of an empty set $\emptyset$, it follows that

$$\begin{aligned} P(S) &= P(S \cup \emptyset) \\ &= P(S) + P(\emptyset) \quad \text{by the third axiom} \end{aligned}$$

and hence $P(\emptyset) = 0$

3. If A and B are sample events in a sample space S and $A \subset B$, then

$$P(A) \leq P(B)$$

Since $A \subset B$, we can write

$$B = A \cup (A' \cap B)$$

Then, since A and $A' \cap B$ are mutually exclusive, we get

$$\begin{aligned} P(B) &= P(A) + P(A' \cap B) \\ &\geq P(A) \end{aligned}$$

Conditional Probability

Since a sample space is by no means always self-evident, it often helps to use the symbol $P(A | S)$ to denote the conditional probability of an event A relative to the sample space S or in other words 'Probability of A given S '.

Conditional Probability: If A and B are any two events in a sample space S and $P(A) \neq 0$, the conditional probability of B given A is

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

Example 4. A bin contains 5 defective (that immediately fail when put in use), 10 partially defective (that fail after a couple of hours of use), and 25 acceptable transistors. A transistor is chosen at random from the bin and put into use. If it does not immediately fail, what is the probability it is acceptable?

Solution: Since the transistor did not immediately fail, we know that it is not one of the 5 defective ones and so the desired probability is:

$$\begin{aligned} P\{ \text{acceptable} \mid \text{not defective} \} &= \frac{P\{ \text{acceptable, not defective} \}}{P\{ \text{not defective} \}} \\ &= \frac{P\{ \text{acceptable} \}}{P\{ \text{not defective} \}} \end{aligned}$$

where the last equality follows since the transistor will be both acceptable and not defective if it is acceptable. Hence, assuming that each of the 40 transistors is equally likely to be chosen, we obtain that

$$P\{ \text{acceptable} \mid \text{not defective} \} = \frac{25/40}{35/40} = \frac{5}{7}$$

It should be noted that we could also have derived this probability by working directly with the reduced sample space. That is, since we know that the chosen transistor is not defective, the problem reduces to computing the probability that a transistor, chosen at random from bin containing 25 acceptable and 10 partially defective transistors, is acceptable. This is clearly equal to $\frac{25}{35}$.

Example 5. Ms. Mwangi figures that there is a 30% chance that her company will set up a branch office in Nairobi. If it does, she is 60% certain that she will be made manager of this new operation. What is the probability that Ms. Mwangi will be a Nairobi branch office manager?

Solution: If we let B denote the event that the company sets up a branch office in Nairobi and M the event that Ms. Mwangi is made the Nairobi branch manager, then the desired probability is $P(BM)$, which is obtained as follows

$$\begin{aligned} P(BM) &= P(B)P(M \mid B) \\ &= (0.3)(0.6) \\ &= 0.18 \end{aligned}$$

There is an 18% chance that Ms. Mwangi will be the Nairobi Manager.

Independent Events

Two event A and B are independent if the occurrence or non-occurrence of either one does not affect the probability of occurrence of the other.

Independence: Two events A and B are independent if any of the following equivalent statements are true:

- (1) $P(B \mid A) = P(B)$
- (2) $P(A \mid B) = P(A)$
- (3) $P(A \cap B) = P(A) \cdot P(B)$

Example 6. An urn contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its color noted then it is replaced. A second ball is selected and its color noted. Find the probability of each of these

- (i) Selecting 2 blue balls

(ii) Selecting 1 blue ball and then a white ball

(iii) Selecting 1 red ball and then 1 blue ball

Solution

(i)

$$\begin{aligned}P(\text{ blue and blue }) &= P(\text{ blue }) \cdot P(\text{ blue }) \\&= \frac{2}{10} \cdot \frac{2}{10} \\&= \frac{4}{100} = \frac{1}{25}\end{aligned}$$

(ii)

$$\begin{aligned}P(\text{ blue and white }) &= P(\text{ blue }) \cdot P(\text{ white }) \\&= \frac{2}{10} \cdot \frac{5}{10} \\&= \frac{10}{100} = \frac{1}{10}\end{aligned}$$

(iii)

$$\begin{aligned}P(\text{ red and blue }) &= P(\text{ red }) \cdot P(\text{ blue }) \\&= \frac{3}{10} \cdot \frac{2}{10} \\&= \frac{6}{100} = \frac{3}{50}\end{aligned}$$

Bayes' Theorem

In some examples, we might know one conditional probability but would like to calculate a different one. For example, from the definition of conditional probability

$$P(A \cap B) = P(A | B)P(B) = P(B \cap A) = P(B | A)P(A)$$

Considering the second and the last terms, we can write

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \quad \text{for } P(B) > 0$$

In general, before we express Bayes' Theorem, we need the rule of total probability

Rule of Total Probability: If the events $B_1, B_2, \dots$, and B_k constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i = 1, 2, 3, \dots, k$, then for any event A in S

$$P(A) = \sum_{i=1}^k P(B_i) \cdot P(A | B_i)$$

Theorem If $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i = 1, 2, \dots, k$, then for any event A in S such that $P(A) \neq 0$

$$P(B_r | A) = \frac{P(B_r) \cdot P(A | B_r)}{\sum_{i=1}^k P(B_i) \cdot P(A | B_i)}$$

for $i = 1, 2, \dots, k$

Proof Writing $P(B_r | A) = \frac{P(A \cap B_r)}{P(A)}$ in accordance with the definition of conditional probability, we have only to substitute $P(B_r) \cdot P(A | B_r)$ for $P(A \cap B_r)$ and the formula for $P(A)$ above.

Example 7. Because a new medical procedure has been shown to be effective in the early detection of an illness, a medical screening of the population is proposed. The probability that the test correctly identifies someone with the illness as positive is 0.99, and the probability that the test correctly identifies someone without the illness as negative is 0.95. The incidence of the illness in the general population is 0.0001. You take the test, and the result is positive. What is the probability that you have the illness?

Solution: Let A denote the event that you have the illness, and let B denote the event that the test signals positive. The probability requested can be denoted as $P(A | B)$. The probability that the test correctly signals someone without the illness as negative is 0.95. Consequently, the probability of a positive test without the illness is

$$P(B | A')$$

From Bayes' Theorem,

$$\begin{aligned} P(A | B) &= \frac{P(B | A) \cdot P(A)}{[P(B | A) \cdot P(A) + P(B | A') \cdot P(A')]} \\ &= \frac{0.99(0.0001)}{[0.99(0.0001) + 0.05(1 - 0.0001)]} \\ &= 1/506 = 0.002 \end{aligned}$$

Example 8. A toy is rejected if the design is faulty or not. The probability that the design is faulty is 0.1 and that the toy is rejected if the design is faulty is 0.95 and otherwise 0.45. If a toy is rejected, what is the probability that it is due to faulty design?

Solution: Let D_1 and D_2 denote the events that the design is faulty or not. Let A denote the event that the toy is rejected. $P(D_1) = 0.1$ and $P(D_2) = 1 - 0.1 = 0.9$ and $P(A | D_1) = 0.95$ and $P(A | D_2) = 0.45$. Therefore, the probability of rejection due to faulty design is

$$\begin{aligned} P(D_1 | A) &= \frac{P(D_1) P(A | D_1)}{P(D_1) \cdot P(A | D_1) + P(D_2) \cdot P(A | D_2)} \\ &= \frac{0.1 \times 0.95}{0.1 \times 0.95 + 0.9 \times 0.45} = 0.19 \end{aligned}$$

The Concept of Random Variables

Overview of the Concept of Random Variable

A random variable is a concept in probability and statistics that assigns numerical values to the outcomes of a random phenomenon. Unlike deterministic variables, which have fixed values, random variables can take different values, each with a certain probability. Random variables are essential for modeling and analyzing situations where there are uncertainty or variability. Understanding random variables allows you to quantify and make predictions about uncertain events, providing you with a firm foundation for more advanced studies in probability, statistics and data analysis.

Definitions

- **Random Experiment** - This is an experiment that may be repeated as often as we want and whose outcome cannot be predicted. Note that the outcomes are different even though the experiment is repeated in the same manner every time. Some typical examples of random experiments include rolling a die, the toss of a coin and drawing a card from a deck among others.
- **Random variables** - this is a numerical outcome of a random experiment. It assigns a real number to each outcome in the sample space of a random experiment.
- **Discrete Random Variable** - These are random variables which take on countable number of distinct values. For example the number of heads in a coin toss, the number of students in a class, and so on.
- **Continuous Random Variable** - These are random variables that take on uncountable number of values. Examples include height, weight, time.

Discrete Random Variables

A random variable X is said to be discrete if it can take only a finite or countable number of possible values x . Consider an experiment of flipping a fair coin 3 times. The number of tails that appear is noted as a discrete random variable, that is,

$$X = \text{Number of tails that appear in 3 flips of a coin.}$$

This is an example of a random experiment and there are 8 possible outcomes of the experiment. Remember the possible outcomes form the sample space. Therefore, this sample space contains the following

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} \quad (1)$$

From this sample space, the corresponding values taken by the random variable are

$$X = \{0, 1, 1, 2, 1, 2, 2, 3\} \quad (2)$$

At this point, you may ask yourself:

What is the probability of X taking a particular value?

To answer this question, we introduce what is called Probability Distribution.

A probability distribution gives the probability of each possible value of the variable under observation. It is similar to frequency distribution. Let us create a probability distribution from the information contained in equations 1 and 2.

- The random variable is denoted by capital X , and the particular value is denoted by small x
- The probability that the variable X take the value x is written as $P(X = x)$ and is called a probability mass function.
- From equation 2, the possible values of X are 4. Therefore,

$$X = \{0, 1, 2, 3\}$$

More specifically.

$$\begin{aligned} \{X = 0\} &= \{HHH\} \\ \{X = 1\} &= \{HHT, HTH, THH\} \\ \{X = 2\} &= \{TTH, HTT, THT\} \\ \{X = 3\} &= \{TTT\} \end{aligned}$$

- The total number of outcomes is 8 . Therefore,

$$\begin{aligned} \{X = 0\} &= \{HHH\} = \frac{1}{8} \\ \{X = 1\} &= \{HHT, HTH, THH\} = \frac{3}{8} \\ \{X = 2\} &= \{TTH, HTT, THT\} = \frac{3}{8} \\ \{X = 3\} &= \{TTT\} = \frac{1}{8} \end{aligned}$$

- The probability distribution becomes

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Remark. The probability that X takes on the value x , $P(X = x)$, is defined as the sum of the probabilities of all points in S that are assigned the value x . We therefore say, that this probability mass function places a mass of $\frac{3}{8}$ on the value $X = 2$. The 'masses' (or probabilities) for a probability mass function should be between 0 and 1. The total mass must add up to 1.

Definition. The probability mass function is a discrete variable or a table, formula or graph that specifies the proportion (or probabilities) associated with each possible value the random variable can take. The mass function $P(X = x) = p(x)$ has the properties

- $0 \leq p(x) \leq 1$
- $\sum_{\text{all } x} p(x) = 1$

Example 9. Two tetrahedral dice are rolled together once and the sum of the scores facing down was noted. Find the probability mass function of the random variable 'the sum of the scores facing down'.

Solution:

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

$X = \{2, 3, 4, 5, 6, 7, 8\}$. The total number of outcomes is 16. The probability mass function is therefore

x	2	3	4	5	6	7	8
$P(X = x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

You can further review the material on terms and rules of probability and the concept of random variable in pages 176-181, 188 - 189 and 244 - 246 from the ebook by Susan Dean and Barbara Illowsky (2018).

Introductory Statistics, 4th edition. OpenStax College.

Click Statistical terms to read the book.

Continuous Random Variables

A continuous random variable assumes any value in an interval on the real line or in a collection of intervals. The sample space for this variable is uncountable. For instance, suppose an experiment involves observing the arrival of cars at a certain period of time along a highway on a particular day. Let T denote the time that lapses before the first arrival, then T is a continuous random variable that assumes values in the interval $[0, \infty)$.

Definition: - A random variable X is continuous if there exists a nonnegative function f so that, for every interval B , the probability that X take a value in

B , that is, $P(X \in B)$ is given as $\int_B f(x)dx$. The function $f = f(x)$ is called a probability density function of X .
Definition: - Let X be a continuous random variable that assumes values in the interval $(-\infty, +\infty)$. Then, $f(x)$ is said to be a probability density function (pdf) of X if it satisfies the following conditions

- $f(x) \geq 0 \forall x$
- $\int_{x=a}^{x=b} f(x)dx = 1$
- $\int_{-\infty}^{+\infty} f(x)dx = 1$

Remark. Note:- The probability of a random variable assuming a value within some given interval from $x = a$ to $x = b$ is defined to be the area under the graph of the probability density function between $x = a$ and $x = b$.

Example 10. Let X be a continuous random variable. Show that the function

$$f(x) = \begin{cases} \frac{1}{2}x, & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

is a pdf of X , hence compute $P(0 \leq X \leq 1)$ and $P(-1 < X < 1)$.

Solution:

- From the expression of $f(x)$ in the example, we see that $f(x) \geq 0$ for all values of x . This satisfies the first property.
- Next we solve, $\int_0^2 \frac{1}{2}x dx$

$$\begin{aligned} \int_0^2 \frac{1}{2}x dx &= \frac{1}{4} [x^2]_0^2 \\ &= 1 - 0 = 1 \end{aligned}$$

This satisfies the second property.

- Therefore $f(x)$ is indeed a pdf of X .

$$\begin{aligned} P(-1 < X < 0) &= P(-1 < X < 0) + P(0 < X < 1) \\ &= 0 + \frac{1}{4} = \frac{1}{4} \end{aligned}$$

Cumulative Distribution Function

Definition:- For any random variable X , we define the cumulative distribution function (CDF), denoted by $F(x)$ as

$$F(x) = P(X \leq x) = \begin{cases} \sum_{t=-\infty}^x f(t), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^x f(t)dt, & \text{if } X \text{ is continuous} \end{cases}$$

for every x .

Properties of any CDF

1. $\lim_{x \rightarrow +\infty} F(x) = 1$ and $\lim_{x \rightarrow -\infty} F(x) = 0$
2. $F(x)$ is a non-decreasing function.
3. $F(x)$ is a right continuous function of x .

Remark. If the CDF is $F(x)$ and the pdf is $f(x)$, then we can differentiate the cdf to get the pdf, and integrate the pdf to get the cdf.

For any random variable X and real values a and b , where $a < b$,

$$P(a \leq X \leq b) = F(b) - F(a)$$

Example 11. Let X be a discrete random variable with pmf given by

$$f(x) = \begin{cases} \frac{1}{20}(1+x) & \text{for } x = 1, 2, 3, 4, 5 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the cdf of X hence compute $P(X > 3)$.

Solution:

$$\begin{aligned} F(x) &= \sum_{t=-\infty}^x f(t) \\ &= \frac{1}{20} \sum_{t=1}^x (x+1) = \frac{1}{20} (2 + 3 + 4 + \cdots + x) \\ &= \frac{1}{20} \left\{ \frac{x}{2} [4 + (x-1)] \right\} \\ &= \frac{x(x+3)}{40} \end{aligned}$$

- We then break it down for various ranges of x

$$F(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{x(x+3)}{40} & \text{for } x = 1, 2, 3, 4, 5 \\ 1 & \text{for } x > 5 \end{cases}$$

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \frac{3 \cdot 6}{40} = \frac{11}{20}$$

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/3NtTAuGSAfc>



Discrete and Continuous Random Variables

You can further review the material on Discrete and Continuous Random Variables in pages 244 - 246 and 312 - 314 from the ebook by Susan Dean and Barbara Illowsky (2018). Introductory Statistics, 4th edition. OpenStax College.

Expectation and Variance of a Random Variable

Expected Values

One of the most important things we'd like to know about a random variable is - what value does it take on average?. This value is found as the average of all possible values, weighted by how often they occur (i.e. probability).

Definition. Definition:- Let X be a random variable with probability distribution $P(X = x)$. Then the expected value of X , denoted $E(X)$ or μ , is given by,

$$E(X) = \mu = \begin{cases} \sum_{x=-\infty}^{\infty} x \cdot p(X = x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot p(X = x) & \text{if } X \text{ is continuous} \end{cases}$$

Definition. Theorem: Let X be a random variable with probability distribution $P(X = x)$ and let $g(x)$ be a real-valued function of X , then the expected value of $g(x)$ is given by

$$E(g(x)) = \begin{cases} \sum_{x=-\infty}^{\infty} g(x) \cdot p(X = x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) \cdot p(X = x) & \text{if } X \text{ is continuous} \end{cases}$$

Theorem: Let X be a random variable with probability distribution $P(X = x)$. Then

1. $E(c) = c$, where c is an arbitrary constant.
2. $E[aX + b] = aE[X] + b = a\mu + b$, where a and b are arbitrary constants.
3. $E[kg(x)] = kE[g(x)]$, where $g(X)$ is a function of X .
4. $E[a \cdot g_1(x) \pm b \cdot g_2(x)] = aE[g_1(x)] \pm bE[g_2(x)]$

Proof:

We start with continuous random variable

$$1. E[c] = \int_{\text{All } x} c \cdot P(X = x)dx = c \cdot \int_{\text{All } x} P(X = x)dx = c \cdot 1 = c$$

$$\begin{aligned} E[aX + b] &= \int_{\text{All } x} (ax + b) \cdot P(X = x)dx \\ &= \int_{\text{All } x} ax \cdot P(X = x)dx + \int_{\text{All } x} b \cdot P(X = x)dx \\ &= a \int_{\text{All } x} x \cdot P(X = x)dx + b \int_{\text{All } x} P(X = x)dx \\ &= aE[x] + b = a \cdot \mu + b \end{aligned}$$

2.

$$\begin{aligned} E[kg(x)] &= \int_{\text{All } x} kg(x) \cdot P(X = x)dx \\ &= k \int_{\text{All } x} g(x)P(X = x)dx \\ &= kE[g(x)] \end{aligned}$$

3.

$$\begin{aligned} E[a \cdot g_1(x) \pm b \cdot g_2(x)] &= \int_{\text{All } x} [E[a \cdot g_1(x) \pm b \cdot g_2(x)]] \cdot P(X = x)dx \\ &= \int_{\text{All } x} ag_1(x) \cdot P(X = x)dx \pm \int_{\text{All } x} bg_2(x) \cdot P(X = x)dx \\ &= E[g_1(x)] \pm E[g_2(x)] \\ &= aE[g_1(x)] \pm bE[g_2(x)] \end{aligned}$$

• For discrete random variable.

$$1. E[X] = \sum_{\text{all } x} c \cdot p(X = x) = c \sum_{\text{all } x} p(X = x) = c(1) = c$$

$$\begin{aligned} E[aX + b] &= \sum_{\text{All } x} (ax + b) \cdot P(X = x) \\ &= \sum_{\text{All } x} ax \cdot P(X = x) + \sum_{\text{All } x} b \cdot P(X = x) \\ &= a \sum_{\text{All } x} x \cdot P(X = x) + b \sum_{\text{All } x} P(X = x) \\ &= aE[x] + b = a \cdot \mu + b \end{aligned}$$

3.

$$\begin{aligned} E[kg(x)] &= \sum_{\text{All } x} kg(x) \cdot P(X = x) \\ &= k \sum_{\text{All } x} g(x)P(X = x) \\ &= kE[g(x)] \end{aligned}$$

4.

$$\begin{aligned}
 E[a \cdot g_1(x) \pm b \cdot g_2(x)] &= \sum_{\text{All } x} [E[a \cdot g_1(x) \pm b \cdot g_2(x)]] \cdot P(X = x) \\
 &= \sum_{\text{All } x} a g_1(x) \cdot P(X = x) \pm \sum_{\text{All } x} b g_2(x) \cdot P(X = x) \\
 &= E[a g_1(x)] \pm E[b g_2(x)] \\
 &= a E[g_1(x)] \pm b E[g_2(x)]
 \end{aligned}$$

Variance and Standard Deviation

Definition. Let X be a random variable with mean $E(X) = \mu$, the variance denoted $\text{Var}(X)$ or σ^2 is given by

$$\text{Var}(x) = \sigma^2 = E[X - \mu]^2$$

The units of variance are square units. The quantity that has correct units is standard deviation, denoted σ . It is actually the positive square root of $\text{Var}(x)$.

Theorem:-

The properties of variance are

1. $\text{Var}(X) = E(X - \mu)^2 = E(X)^2 - \mu^2$
2. $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Proof: -

1.

$$\begin{aligned}
 \text{Var}(X) &= E(X - \mu)^2 \\
 &= E(X^2 - 2X\mu + \mu^2) \\
 &= E(X)^2 - 2\mu E(X) + \mu^2 \\
 &= E(X)^2 - \mu^2
 \end{aligned}$$

2.

$$\begin{aligned}
 \text{Var}(aX + b) &= E[(aX + b) - (a\mu + b)]^2 \\
 &= E[a(X - \mu)]^2 \\
 &= E[a^2(X - \mu)^2] \\
 &= a^2 E[(X - \mu)^2] \\
 &= a^2 \cdot \text{Var}(X)
 \end{aligned}$$

Example 12. Given a probability distribution of X as shown below, find the mean and standard deviation of X .

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$

Solution:

- We begin by creating a table with all the information we need, that is, $E(X) = xp(X = x)$ and $x^2p(X = x)$

x	1	1	2	3	Total
$p(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	1
$xp(X = x)$	0	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{7}{4}$
$x^2p(X = x)$	0	$\frac{1}{4}$	$\frac{3}{2}$	$\frac{9}{4}$	4

•

$$E(X) = \mu$$

$$= 1.75 = \sum_{x=0}^3 xp(X = x)$$

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$\sigma^2 = 4 - 1.75^2$$

$$= 0.9375$$

$$\sigma = 0.968246$$

The standard deviation is therefore 0.968246.

Example 13. A continuous random variable X has a pdf given by

$$f(x) = \begin{cases} \frac{1}{2}x & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the mean and standard deviation of X .

Solution:

- We start by finding both $E(X)$ and $E(X^2)$.

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$= \int_0^2 \frac{1}{2} \cdot x^2 dx$$

$$= \left[\frac{x^3}{6} \right]_0^2 = \frac{4}{3}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$$

$$= \int_0^2 \frac{1}{2} \cdot x^3 dx$$

$$= \left[\frac{x^4}{8} \right]_0^2 = 2$$

- The mean is $E(X) = \frac{4}{3}$.
- We then solve for the variance and standard deviation

$$\begin{aligned}\text{Var}(X) &= E(X^2) - (E(X))^2 \\ \sigma^2 &= 2 - \left(\frac{4}{3}\right)^2 \\ \sigma &= \sqrt{2 - \left(\frac{4}{3}\right)^2} \\ &= \frac{\sqrt{2}}{3}\end{aligned}$$

- the standard deviation is therefore $\frac{\sqrt{2}}{3}$

To learn more about expectation and variance of a random variable, watch the video below. **Video** Visit the URL below to view a video:

<https://www.youtube.com/embed/D8yUPdUdnzw>



Expectation and Variance of a Random Variable